

Creating Maven Ansible Web App:

Step 1: This below Maven command creates a new ansible web application project named **MavenAnsibleWebApp** using the **maven-archetype-webapp** template. It sets the group ID to com.example and runs in non-interactive mode, generating all project files automatically.

```
mvn archetype:generate -DgroupId=com.example -DartifactId=MavenAnsibleWebApp -DarchetypeArtifactId=maven-archetype-webapp -DinteractiveMode=false
```



```
vboxuser@ubuntu220451tsvmo:~$ cd Devops
vboxuser@ubuntu220451tsvmo:~/Devops$ cd shubham
vboxuser@ubuntu220451tsvmo:~/Devops/shubham$ mvn archetype:generate \
-DgroupId=com.example \
-DartifactId=MavenAnsibleWebApp1 \
-DarchetypeArtifactId=maven-archetype-webapp \
-DinteractiveMode=false
[INFO] Scanning for projects...
[INFO] ----- org.apache.maven:standalone-pom -----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
Downloading from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml
Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (18 MB at 3.1 MB/s)
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-webapp:1.0
[INFO] -----
[INFO] Parameter: basedir, Value: /home/vboxuser/Devops/shubham
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MavenAnsibleWebApp1
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/vboxuser/Devops/shubham/MavenAnsibleWebApp1
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 17.440 s
[INFO] Finished at: 2026-05-10T19:07:02+05:30
[INFO] -----
vboxuser@ubuntu220451tsvmo:~/Devops/shubham$
```

Step 2: Configure Maven Ansible Web App Project (pom.xml)

In this step, we move to the project directory using `cd MavenAnsibleWebApp` and open the Maven configuration file using `gedit pom.xml`. Then, we write the required project details, dependencies, WAR packaging settings, and build configuration for creating the web application.

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
```

```
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
```

```
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-
v4_0_0.xsd">
```

```
<modelVersion>4.0.0</modelVersion>
```

```
<groupId>com.example</groupId>
```

<artifactId>MavenAnsibleWebApp1</artifactId>

<packaging>war</packaging>

<version>1.0-SNAPSHOT</version>

<name>MavenAnsibleWebApp1</name>

<url>http://maven.apache.org</url>

<dependencies>

<dependency>

<groupId>junit</groupId>

<artifactId>junit</artifactId>

<version>3.8.1</version>

<scope>test</scope>

</dependency>

<!-- Servlet API for Java EE -->

<dependency>

<groupId>javax.servlet</groupId>

<artifactId>javax.servlet-api</artifactId>

<version>4.0.1</version>

<scope>provided</scope>

</dependency>

</dependencies>

<build>

<plugins>

<plugin>

<groupId>org.apache.maven.plugins</groupId>

<artifactId>maven-war-plugin</artifactId>

<version>3.4.0</version>

<!-- Update to latest stable version -->

</plugin>

</plugins>

<finalName>MavenAnsibleWebApp1</finalName>

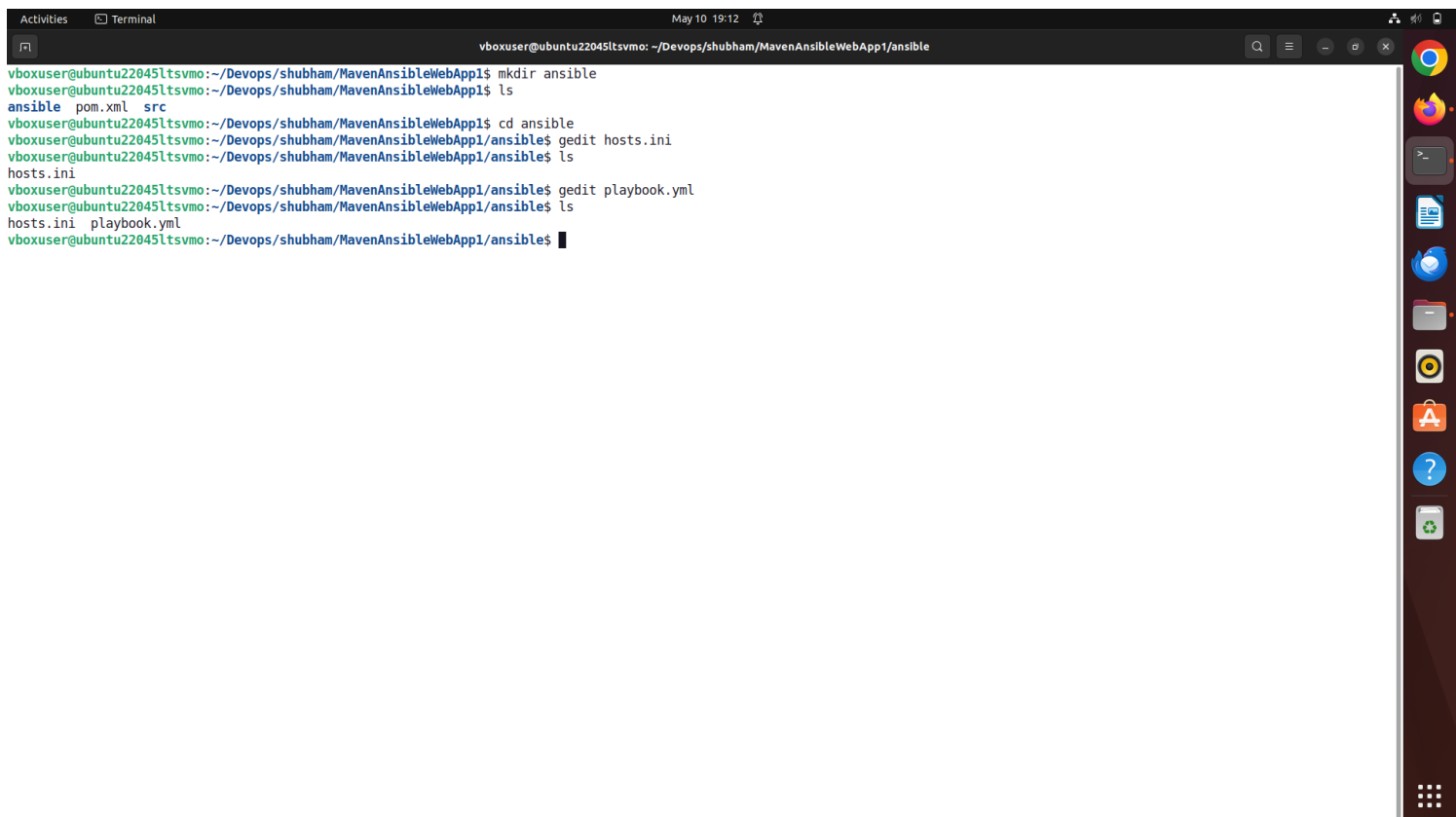
<!-- create a web page with the name MavenAnsibleWebApp, if we want some other name we can change it -->

</build>

</project>

Step 3: Create Ansible Configuration Files

In this step, we create a new folder named ansible inside the project directory using mkdir ansible. Then, we move into the folder using cd ansible and create the required Ansible files using gedit hosts.ini and gedit playbook.yml. These files are used to define target hosts and deployment tasks for automation.

A terminal window screenshot showing the steps to create the ansible directory and files. The terminal is titled 'Terminal' and shows the following commands and output:

```
vboxuser@ubuntu220451tsvmo: ~/Devops/shubham/MavenAnsibleWebApp1/ansible
vboxuser@ubuntu220451tsvmo:~/Devops/shubham/MavenAnsibleWebApp1$ mkdir ansible
vboxuser@ubuntu220451tsvmo:~/Devops/shubham/MavenAnsibleWebApp1$ ls
ansible  pom.xml  src
vboxuser@ubuntu220451tsvmo:~/Devops/shubham/MavenAnsibleWebApp1$ cd ansible
vboxuser@ubuntu220451tsvmo:~/Devops/shubham/MavenAnsibleWebApp1/ansible$ gedit hosts.ini
vboxuser@ubuntu220451tsvmo:~/Devops/shubham/MavenAnsibleWebApp1/ansible$ ls
hosts.ini
vboxuser@ubuntu220451tsvmo:~/Devops/shubham/MavenAnsibleWebApp1/ansible$ gedit playbook.yml
vboxuser@ubuntu220451tsvmo:~/Devops/shubham/MavenAnsibleWebApp1/ansible$ ls
hosts.ini  playbook.yml
vboxuser@ubuntu220451tsvmo:~/Devops/shubham/MavenAnsibleWebApp1/ansible$
```

gedit host.ini

This command is written inside hosts.ini file to define the Ansible inventory. It creates a group named web and adds localhost as the target system, using local connection and Python 3 interpreter for executing Ansible tasks.

```
[web]
```

```
localhost ansible_connection=local ansible_python_interpreter=/usr/bin/python3
```

gedit playbook.yml

This code is written inside playbook.yml file to automate deployment using Ansible. It targets the web host group, copies the generated WAR file from the Jenkins workspace to the Tomcat webapps folder, and then restarts the Tomcat service so the updated application runs successfully.

```
- name: Deploy to Tomcat
```

```
  hosts: web
```

```
  become: yes
```

```
  tasks:
```

```
    - name: Copy WAR file to Tomcat
```

```
      copy:
```

```
        src: "/var/snap/jenkins/5013/workspace/MavenAnsibleWebApp1-CICD/target/MavenAnsibleWebApp1.war"
```

```
        dest: "/opt/tomcat/webapps/MavenAnsibleWebApp1.war"
```

```
        remote_src: no
```

```
    - name: Restart Tomcat
```

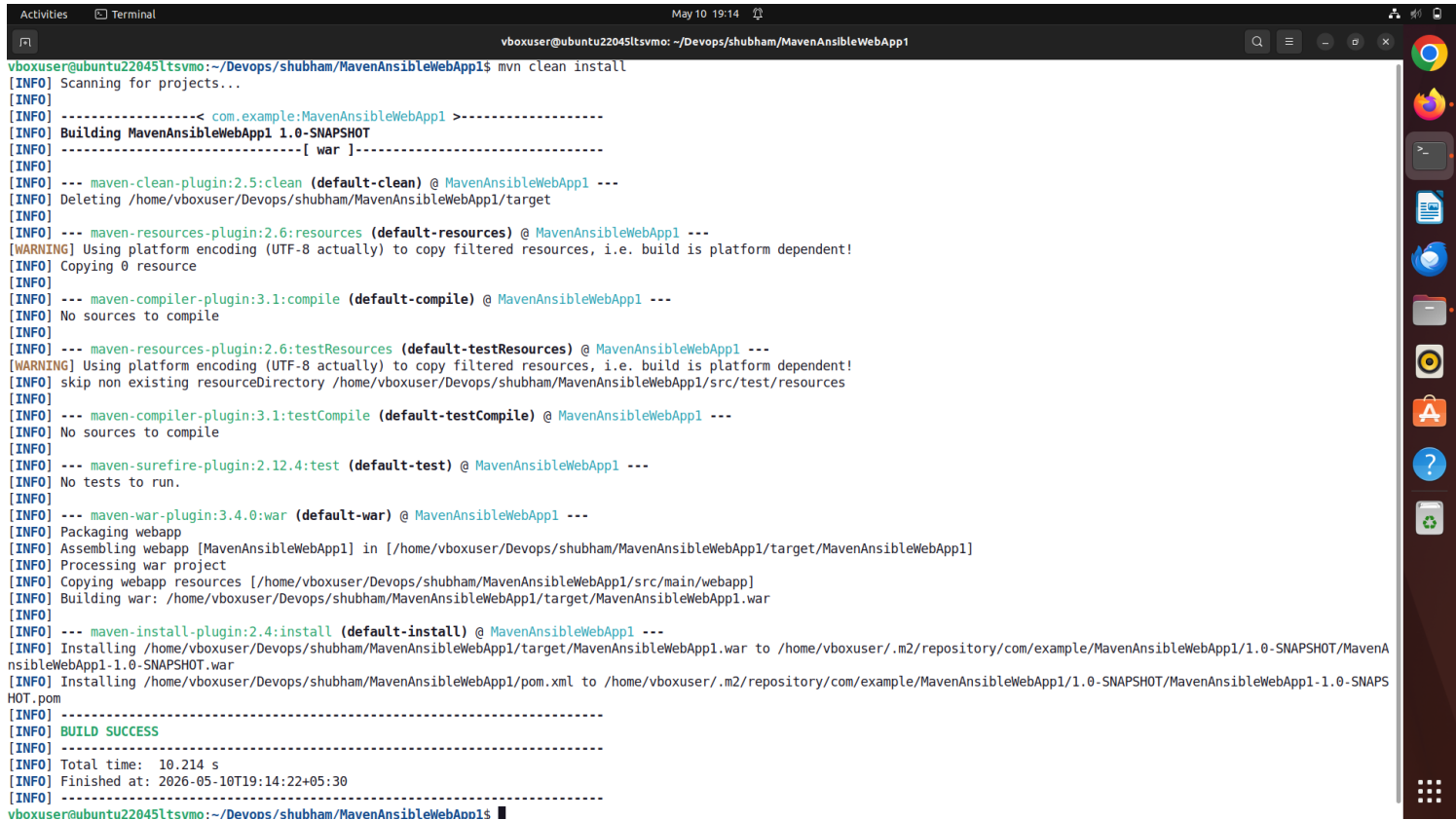
```
      systemd:
```

```
        name: tomcat
```

```
        state: restarted
```

Step 4: Build the Maven Ansible Web Application

In this step, we run the command `mvn clean install` to clean previous files and build the project. Maven compiles the application, packages it as a `.war` file, and stores it in the target folder. After successful completion, the build shows **BUILD SUCCESS** and installs the project files into the local Maven repository.



```
vboxuser@ubuntu220451tsvm0: ~/Devops/shubham/MavenAnsibleWebApp1
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:MavenAnsibleWebApp1 >-----
[INFO] Building MavenAnsibleWebApp1 1.0-SNAPSHOT
[INFO] -----[ war ]-----
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MavenAnsibleWebApp1 ---
[INFO] Deleting /home/vboxuser/Devops/shubham/MavenAnsibleWebApp1/target
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MavenAnsibleWebApp1 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] Copying 0 resource
[INFO]
[INFO] --- maven-compiler-plugin:3.1:compile (default-compile) @ MavenAnsibleWebApp1 ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MavenAnsibleWebApp1 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/vboxuser/Devops/shubham/MavenAnsibleWebApp1/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.1:testCompile (default-testCompile) @ MavenAnsibleWebApp1 ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-surefire-plugin:2.12.4:test (default-test) @ MavenAnsibleWebApp1 ---
[INFO] No tests to run.
[INFO]
[INFO] --- maven-war-plugin:3.4.0:war (default-war) @ MavenAnsibleWebApp1 ---
[INFO] Packaging webapp
[INFO] Assembling webapp [MavenAnsibleWebApp1] in [/home/vboxuser/Devops/shubham/MavenAnsibleWebApp1/target/MavenAnsibleWebApp1]
[INFO] Processing war project
[INFO] Copying webapp resources [/home/vboxuser/Devops/shubham/MavenAnsibleWebApp1/src/main/webapp]
[INFO] Building war: /home/vboxuser/Devops/shubham/MavenAnsibleWebApp1/target/MavenAnsibleWebApp1.war
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MavenAnsibleWebApp1 ---
[INFO] Installing /home/vboxuser/Devops/shubham/MavenAnsibleWebApp1/target/MavenAnsibleWebApp1.war to /home/vboxuser/.m2/repository/com/example/MavenAnsibleWebApp1/1.0-SNAPSHOT/MavenAnsibleWebApp1-1.0-SNAPSHOT.war
[INFO] Installing /home/vboxuser/Devops/shubham/MavenAnsibleWebApp1/pom.xml to /home/vboxuser/.m2/repository/com/example/MavenAnsibleWebApp1/1.0-SNAPSHOT/MavenAnsibleWebApp1-1.0-SNAPSHOT.pom
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 10.214 s
[INFO] Finished at: 2026-05-10T19:14:22+05:30
[INFO] -----
vboxuser@ubuntu220451tsvm0: ~/Devops/shubham/MavenAnsibleWebApp1$
```

Step 5: Verify Project Structure

In this step, we run the `tree` command to display the complete project structure. It shows important files and folders such as `pom.xml`, `Jenkinsfile`, ansible configuration files, source code folders, and the target directory containing the generated `MavenAnsibleWebApp1.war` file ready for deployment.

```
Activities Terminal May 10 19:15
vboxuser@ubuntu220451tsvm0: ~/Devops/shubham/MavenAnsibleWebApp1
[INFO] Assembling webapp [MavenAnsibleWebApp1] in [/home/vboxuser/Devops/shubham/MavenAnsibleWebApp1/target/MavenAnsibleWebApp1]
[INFO] Processing war project
[INFO] Copying webapp resources [/home/vboxuser/Devops/shubham/MavenAnsibleWebApp1/src/main/webapp]
[INFO] Building war: /home/vboxuser/Devops/shubham/MavenAnsibleWebApp1/target/MavenAnsibleWebApp1.war
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MavenAnsibleWebApp1 ---
[INFO] Installing /home/vboxuser/Devops/shubham/MavenAnsibleWebApp1/target/MavenAnsibleWebApp1.war to /home/vboxuser/.m2/repository/com/example/MavenAnsibleWebApp1/1.0-SNAPSHOT/MavenAnsibleWebApp1-1.0-SNAPSHOT.war
[INFO] Installing /home/vboxuser/Devops/shubham/MavenAnsibleWebApp1/pom.xml to /home/vboxuser/.m2/repository/com/example/MavenAnsibleWebApp1/1.0-SNAPSHOT/MavenAnsibleWebApp1-1.0-SNAPSHOT.pom
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 10.214 s
[INFO] Finished at: 2026-05-10T19:14:22+05:30
[INFO]
vboxuser@ubuntu220451tsvm0:~/Devops/shubham/MavenAnsibleWebApp1$ tree
.
├── ansible
│   ├── hosts.ini
│   └── playbook.yml
├── Jenkinsfile
├── pom.xml
├── README.md
├── src
│   └── main
│       ├── resources
│       │   ├── webapp
│       │   │   ├── index.jsp
│       │   │   └── WEB-INF
│       │   │       └── web.xml
│       └── web.xml
└── target
    ├── classes
    ├── MavenAnsibleWebApp1
    │   ├── index.jsp
    │   ├── META-INF
    │   ├── WEB-INF
    │   │   ├── classes
    │   │   │   ├── web.xml
    │   │   └── MavenAnsibleWebApp1.war
    │   └── maven-archiver
    │       └── pom.properties
    └── pom.properties

13 directories, 11 files
vboxuser@ubuntu220451tsvm0:~/Devops/shubham/MavenAnsibleWebApp1$
```

Setting Up Jenkins Pipeline for MavenAnsibleWebApp

Step 1: To deploy **MavenAnsibleWebApp**, open the terminal, cd into the **MavenAnsibleWebApp** folder, create a **Jenkinsfile** containing the pipeline code, and then commit and push it to GitHub.

```
NK@NumanKhan:~/Devops/MavenAnsibleWebApp$ gedit Jenkinsfile
NK@NumanKhan:~/Devops/MavenAnsibleWebApp$ ls
Jenkinsfile pom.xml src
```

pipeline {

agent any // Use any available agent

environment {

LANG = 'en_US.UTF-8'

LC_ALL = 'en_US.UTF-8'

} // this has to be added only if you get an error saying UTF required is 8 but showing in ISO00009

tools {

```
maven 'Maven' // Ensure this matches the name configured in Jenkins
}

stages {

    stage('Checkout') {

        steps {

            git branch: 'master', url: 'https://github.com/ShruthiBGowda/MavenAnsibleWebApp1-
CICD.git'

        }

    }

    stage('Build') {

        steps {

            sh 'mvn clean package' // Run Maven build

        }

    }

    stage('Archive') {

        steps {

            archiveArtifacts artifacts: 'target/*.war', fingerprint:true

        }

    }

    stage('Deploy') {

        steps {

            sh 'mvn clean package'

            sh 'ansible-playbook ansible/playbook.yml -i ansible/hosts.ini'

        }

    }

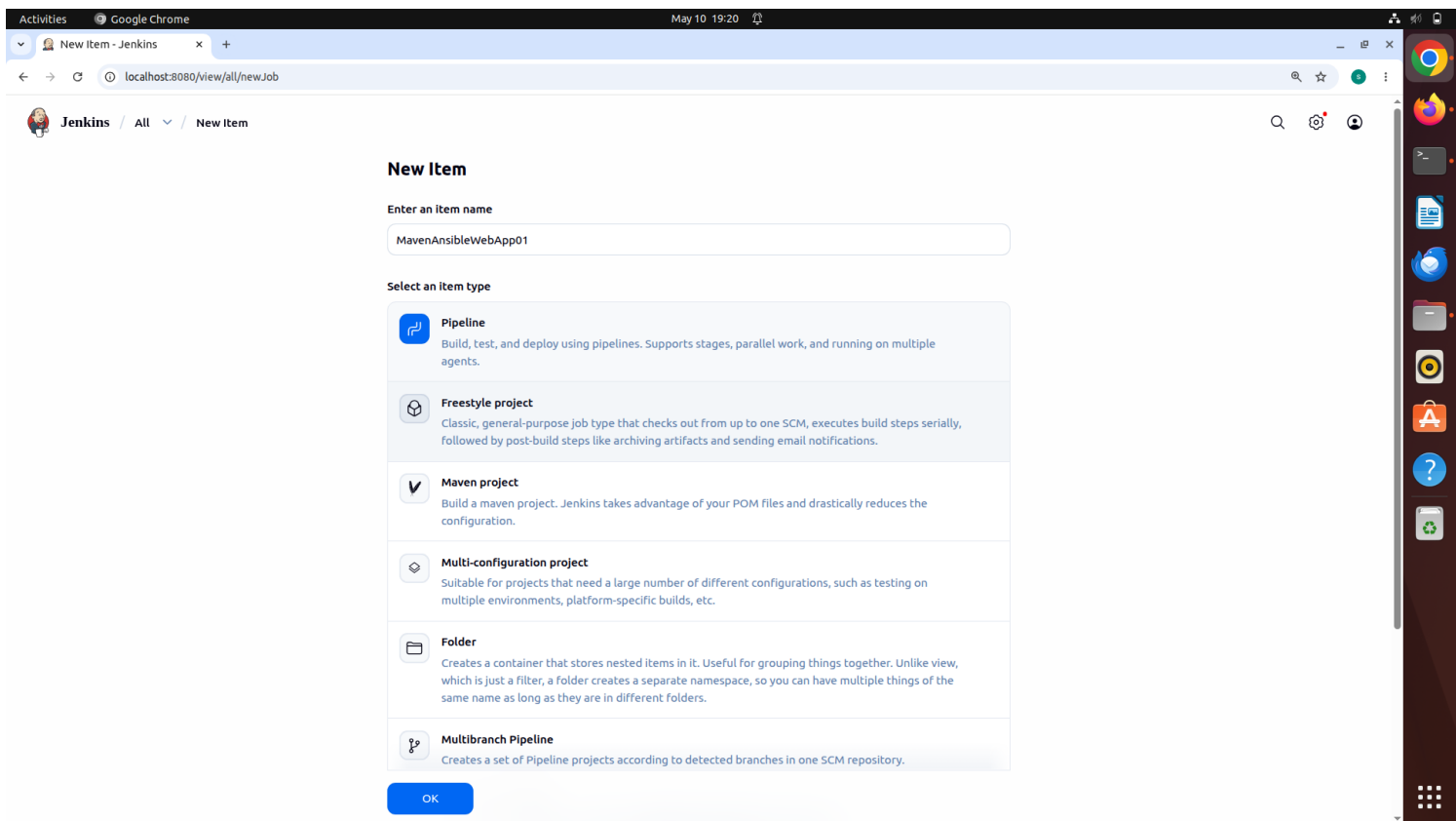
}
```

}

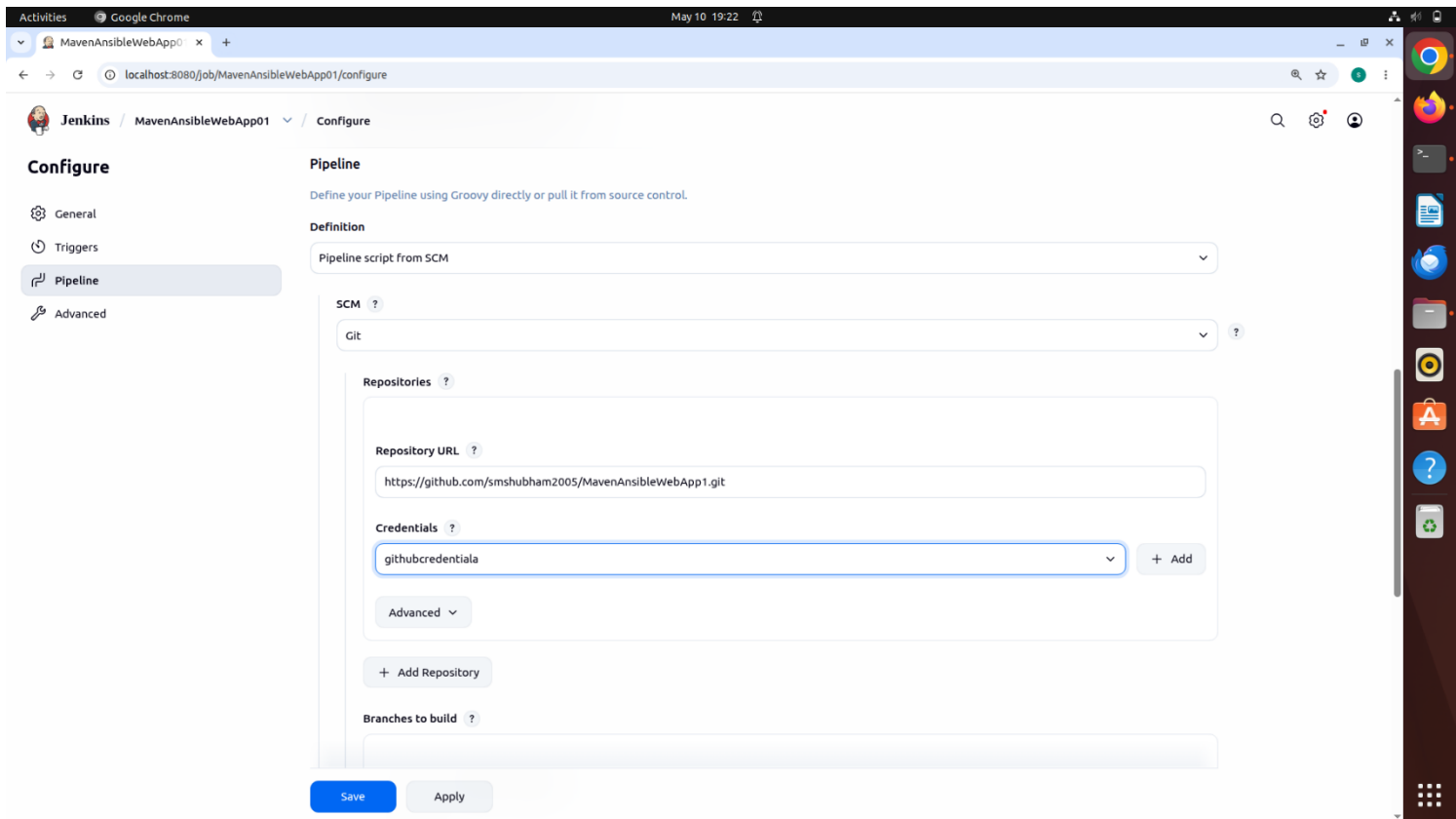
}

}

Step 2: To deploy **MavenAnsibleWebApp**, you first need to create a Jenkins pipeline job that will use the Jenkinsfile from your project repository. Then, go to the Jenkins home page and click on “**New Item**” to create a job, enter the item name (e.g., *MavenAnsibleWebApp*), and select **Pipeline** as the project type. This option allows you to build, test, and deploy applications using a Jenkinsfile. After selecting the pipeline type, click “**OK**” to proceed to the job configuration page.

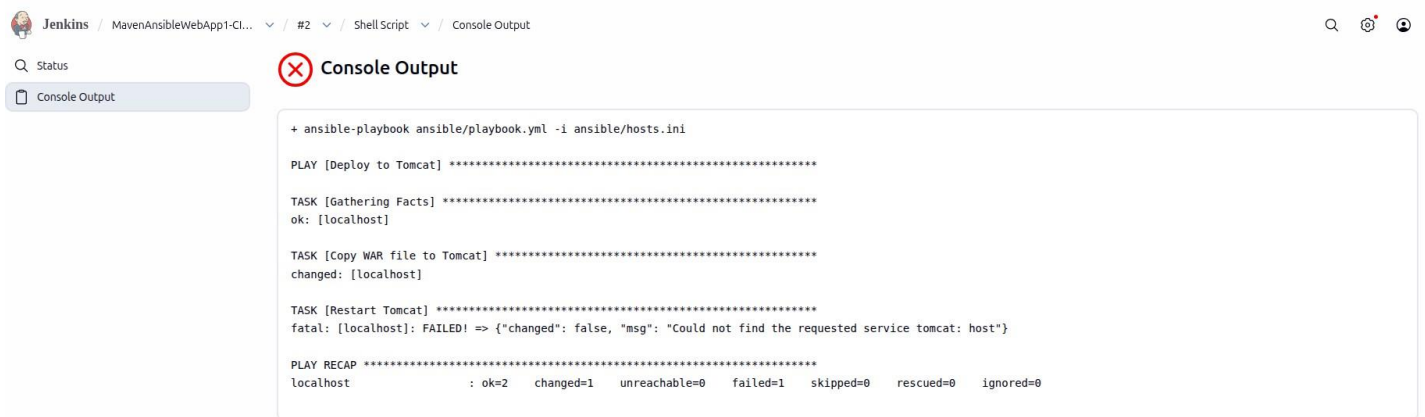


Step 3: In **MavenAnsibleWebApp**, go to the **Pipeline** section and under **Definition**, select **Pipeline script from SCM**. Choose **Git** as the SCM, and under **Repository URL**, paste the GitHub URL of **MavenAnsibleWebApp**. In the **Credentials** field, select the previously created credential (e.g., *GitHub Credential*). Finally, in **Script Path**, enter the name of the Jenkinsfile you created in the MavenAnsibleWebApp folder and pushed to GitHub. Then click **Apply** and **Save** to finalize the configuration.



Step 4: Build and Troubleshoot Deployment in Jenkins

After clicking **Apply and Save**, go back to MavenAnsibleWebApp1-CICD in Jenkins and click **Build Now**. During the initial build, the deployment failed while restarting the Tomcat service because Ansible could not find the service name tomcat.



To resolve this issue, the `playbook.yml` file was updated by replacing the `systemd` restart method with Tomcat manual control commands. Instead of restarting a service named `tomcat`, the script uses `/opt/tomcat/bin/shutdown.sh` to stop the Tomcat server, waits for a few seconds using `sleep 5`, and then starts it again using `/opt/tomcat/bin/startup.sh`. This method works because Tomcat

was installed manually and not registered as a system service. After saving the updated playbook, the Jenkins job was executed again, and the application was deployed successfully without any errors.

The below is the updated playbook.yml file, modified to resolve the Tomcat restart issue by using shutdown and startup commands instead of the systemd service method.

```
- name: Deploy to Tomcat
```

```
  hosts: web
```

```
  become: yes
```

```
  tasks:
```

```
    - name: Copy WAR file to Tomcat
```

```
      copy:
```

```
        src: "/var/snap/jenkins/5013/workspace/MavenAnsibleWebApp1-CICD/target/MavenAnsibleWebApp1.war"
```

```
        dest: "/opt/tomcat/webapps/MavenAnsibleWebApp1.war"
```

```
        remote_src: no
```

```
    - name: Restart Tomcat
```

```
      shell: |
```

```
        /opt/tomcat/bin/shutdown.sh || true
```

```
        sleep 5
```

```
        /opt/tomcat/bin/startup.sh
```

Step 5: After making the changes, go back to **MavenAnsibleWebApp1-CICD** in Jenkins and click **Build Now**. The build should complete successfully without any errors.

Jenkins / MavenAnsibleWebApp1

Status

Changes

Build Now

Delete Pipeline

Configure

Rename

Full Stage View

Stages

Pipeline Syntax

Builds

Filter

Today

- #9 7:23 PM
- #8 4:01 PM
- #7 3:59 PM
- #6 3:53 PM
- #5 3:44 PM

April 15, 2026

- #4 12:48 PM
- #3 9:59 AM
- #2 9:57 AM

localhost:8080/job/MavenAnsibleWebApp1/build?delay=0sec

MavenAnsibleWebApp1

Last Successful Artifacts

MavenAnsibleWebApp1.war 1.82 KIB view

Stage View

	Declarative: Checkout SCM	Declarative: Tool Install	Checkout	Build	Archive	Deploy
Average stage times: (full run time: ~1min 24s)	3s	873ms	3s	20s	1s	30s
#9 May 10 19:23 No Changes	3s	1s	3s	31s	1s	44s
#8 May 10 16:01 1 commit	2s	507ms	2s	16s	1s	30s
#7 May 10 15:59 1 commit	2s	581ms	2s	15s	1s	17s failed
#6 May 10 15:53 No Changes	1s	538ms	3s	16s	2s	25s failed
#5 May 10 15:44 No Changes	4s	1s	3s	23s	1s	34s failed
#4 Apr 15 12:48 No Changes	3s	1s	3s	26s	1s	36s failed

Verify MavenAnsibleWebApp Deployment on Tomcat

After completing the Tomcat installation and configuration, we open the terminal and switch to root user mode using `sudo -i`. Then, we move to the Tomcat binary directory using `cd /opt/tomcat/bin` and start the server using `sh startup.sh`. After Tomcat starts successfully, we open the browser and access `http://localhost:9090/MavenAnsibleWebApp1` to verify the deployed application output.

Activities

Google Chrome

May 10 19:28

MavenAnsibleWebApp1

/manager

localhost:9090/manager/html



Tomcat Web Application Manager

Message: OK

Manager

List ApplicationsHTML Manager HelpManager HelpServer Status

Applications

Path	Version	Display Name	Running	Sessions	Commands
/	None specified	Welcome to Tomcat	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/MavenAnsibleWebApp1	None specified	Archetype Created Web Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/MyMavenWebApp	None specified	Archetype Created Web Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/docs	None specified	Tomcat Documentation	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/examples	None specified	Servlet and JSP Examples	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/host-manager	None specified	Tomcat Host Manager Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/manager	None specified	Tomcat Manager Application	true	2	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes

Deploy

Deploy directory or WAR file located on server

Context Path:

Version (for parallel deployment):

Activities

Google Chrome

May 10 19:38

MavenAnsibleWebApp1

localhost:9090/MavenAnsibleWebApp1/

localhost:9090/MavenAnsibleWebApp1/

shubham

1BI23CS200